

Appendix for Embody4D: A Generalist 4D World Model for Embodied AI

A Real-world Embodied Experiments

In our real-world experiments, the input camera pose is illustrated in Fig. 2. The target output pose is defined by a 20° rightward rotation and a 30 cm forward translation relative to the input. We selected TrajCrafter [9] and ReCamMaster [2], the two top-performing models as baselines to generate videos from identical target viewpoints for a fair comparison. Our robotic manipulation setup employs a third-person view coupled with a wrist-mounted camera as the primary baseline. To evaluate performance improvements, we integrated novel-view data generated by Embody4D, TrajCrafter, and ReCamMaster into the training pipeline and measured their respective performance gains relative to the baseline. Consistent with the main text, our downstream model is the current state-of-the-art Vision-Language-Action model $\pi_{0.5}$. In addition, we collect one hundred human demonstration trajectories for training.

The experimental results are summarized in Tab. 1. ReCamMaster, designed for continuous camera motion via direct extrinsic control, struggles with motion instability when simulating the fixed camera poses required by our downstream tasks, leading to a performance decline. TrajCrafter also delivers sub-optimal performance in embodied scenarios, often suffering from visual artifacts and hallucinations, resulting in marginal improvements. In contrast, as shown in Fig. 1, Embody4D achieves superior performance, reaching a task success rate of 74% and demonstrating exceptional out-of-distribution (OOD) generalization. Compared to the experiments in the main text that omit the wrist-view, these results prove that even when the model relies heavily on the wrist-mounted camera, the addition of an enhanced third-person view remains crucial. This further validates that our novel-view generation approach can serve as an effective data engine for downstream tasks, providing a new paradigm for 4D world models in embodied AI, particularly demonstrating strong generalization capabilities on OOD tasks. Additionally, our method supports scalability by integrating multiple third-person image sequences into the VGGT [7] pipeline, enabling a more comprehensive point cloud reconstruction that can be projected onto arbitrary target views for completion and novel view synthesis.

B More Experimental Details

B.1 Dataset Curation

When rendering the robotic arm foreground in the MuJoCo [6] simulation environment, we ensure that the simulated motion closely mimics real world dynamics by strictly constraining the movement magnitude. Specifically, each movement step size is set to one hundredth of the total range of motion for each joint.

Fig. 1: Comparison of successful novel-view synthesis counts
under four experimental settings on seen tasks and OOD unseen tasks

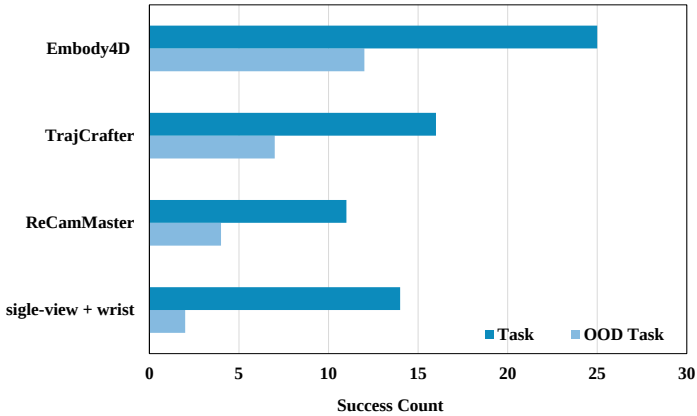


Table 1: Real-world embodied experiments on FR3 with $\pi_{0.5}$ Droid. Each task has 10 trials with randomized object positions. Tasks: T1-T3 (seen), T4-T5 (unseen).

Task	Baselines			Ours Embod4D
	Single-view with wrist	ReCamMaster [2]	TrajCrafter [9]	
T1 (Grapes→Bowl)	5/10	4/10	6/10	8/10
T2 (Grapes→Plate)	5/10	5/10	6/10	8/10
T3 (Mangoes→Bowl)	4/10	2/10	4/10	9/10
T4 (Lemons→Bowl)	1/10	2/10	0/10	6/10
T5 (Bananas→Plate)	1/10	2/10	7/10	6/10
Success Rate (SR)	32%	30%	46%	74%

To maintain spatial consistency, the camera placement within the simulation environment must strictly adhere to the extrinsic parameters of the target backgrounds intended for compositing. This process necessitates a precise coordinate transformation to map the OpenGL coordinate system used in the DL3DV [5] dataset into the XML format required by MuJoCo, which is defined by specific values for distance as well as azimuth and elevation angles.

Our empirical evaluation demonstrated that employing 49 frame sequences with independent camera and background poses leads to suboptimal performance. The compounding motion of both the foreground robotic arm and the background camera generates highly complex data that hinders the ability of the model to achieve convergence. To mitigate this issue, each data sample consists of two separate videos captured from two distinct and fixed camera poses. To prevent excessive parallax which might cause the robotic arm to drift out of the

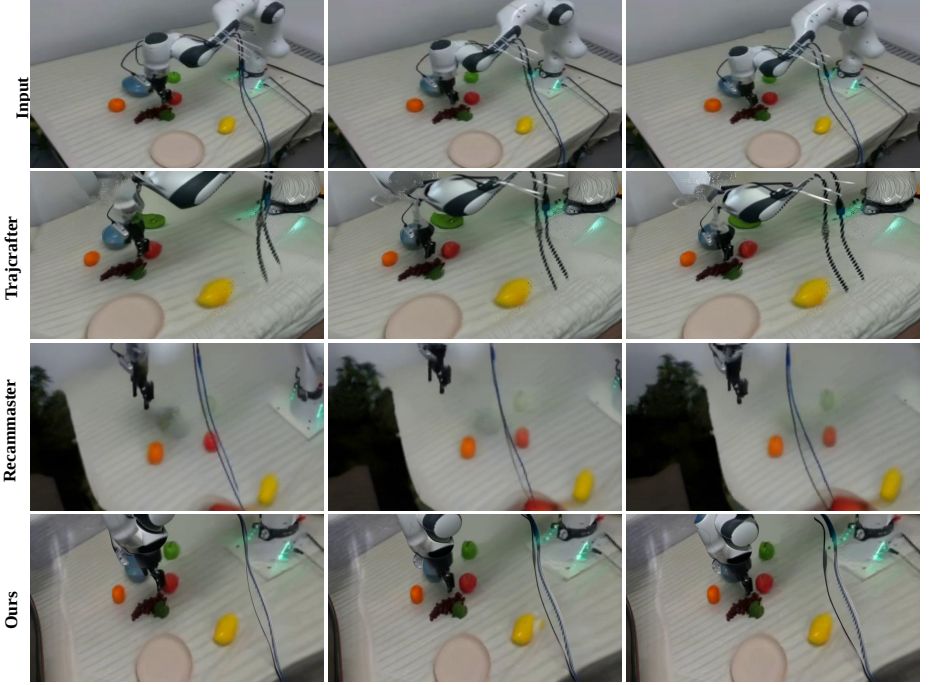


Fig. 2: Visualization results of real-world embodied experiments.

frame, we utilize the GPT-4o [1] model to evaluate and select background pairs that maintain a moderate level of parallax.

As illustrated in Fig. 3, our training pairs are annotated by concatenating the identifier of the robotic arm with a detailed background description to form the final text prompt. This composition based approach yields more precise labels than the direct annotation of the final composited video sequence. For this purpose, we employ the CogVLM2 Llama3 Caption model [8] as our primary labeling tool to ensure high quality descriptive metadata.

B.2 Acquisition of Interaction Priors

When utilizing the interaction-aware attention mechanism, it is first necessary to obtain a motion prior bias. For data captured in real world scenarios, camera movement often introduces a certain degree of jitter, which in turn leads to noise in the optical flow estimation [4]. In cases where the noise level is excessively high, we directly employ the Segment Anything Model 3 [3] to perform segmentation. The input prompt for this process is defined as “robot arm and objects”. As illustrated in Fig. 4, we provide several examples of the resulting optical flow estimations alongside the segmentation masks generated by the Segment Anything Model 3, all masks will be normalized to the range of 0 to 1 for calculations.

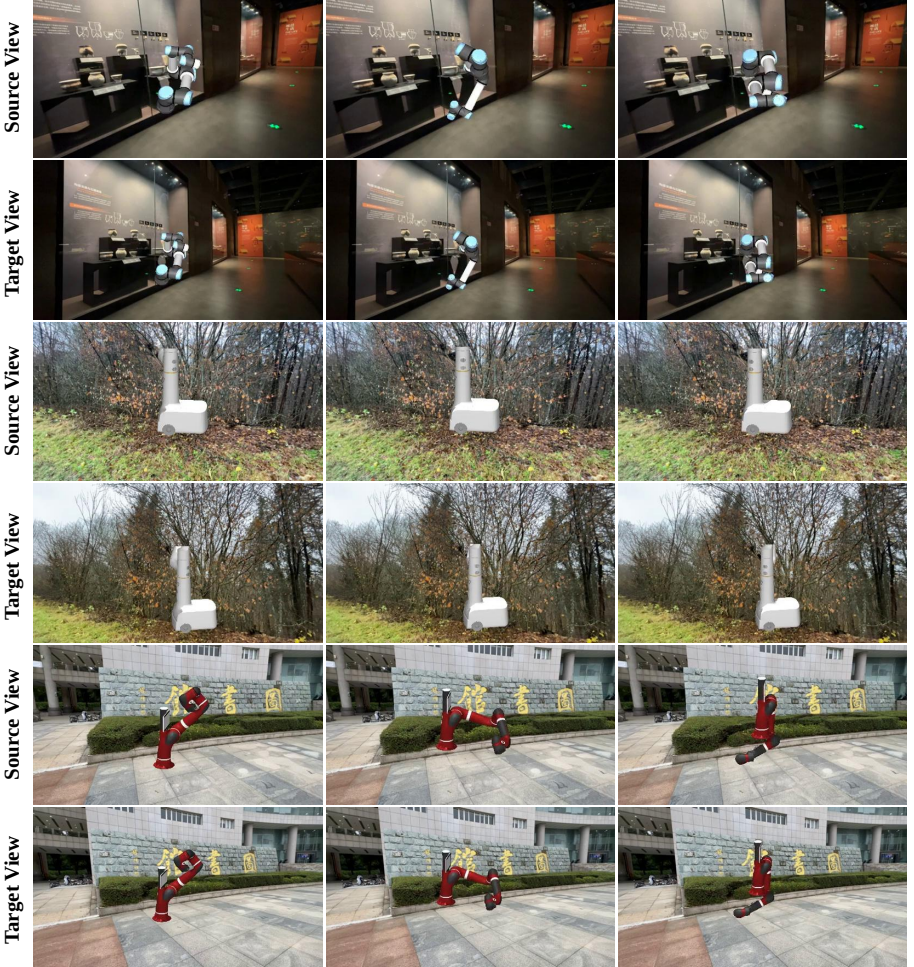


Fig. 3: Visualization results of compositional 4D embodied data, featuring 4D-consistent sequences with diverse robotic arm foregrounds and varied backgrounds.

B.3 Training Implementation Details

We implement our Embody4D using the Wan2.1-T2V-1.3B diffusion transformer as the generative backbone. All parameters are fully fine-tuned in an end-to-end manner. During training, the frame resolution is fixed at 384×672 , with a sequence length of 49 frames.

We adopt a two-stage training strategy. In the first stage, the model is trained on our self-constructed combinatorial real-world 4D dataset to enhance temporal consistency. In the second stage, we freeze the interaction-aware blocks and utilize pseudo-4D data, generated by projecting interaction data onto arbitrary perspectives and back-projecting them to the initial view. This augmentation

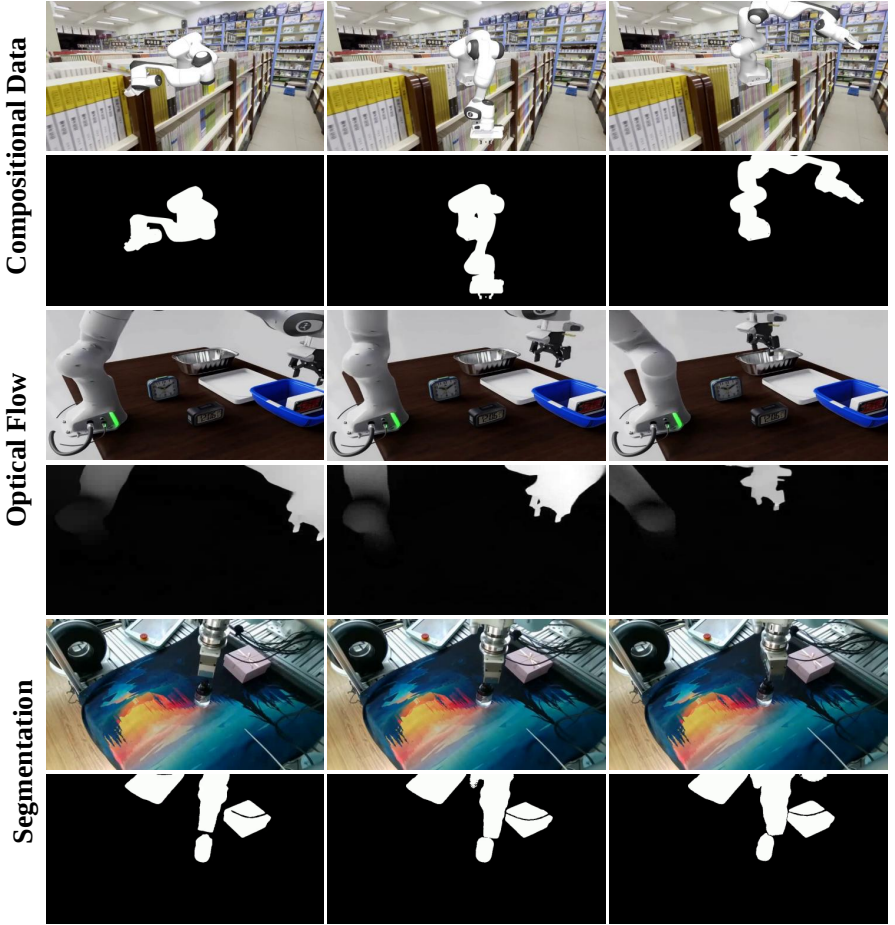


Fig. 4: Visualization of interaction motion priors. Showcasing three different foreground acquisition strategies: direct acquisition from compositional data, optical flow estimation, and video object segmentation.

enables the model to generalize effectively to complex real-world interactions. Finally, all model parameters are fully fine-tuned in an end-to-end manner.

The training configurations are summarized as follows:

- **Stage 1:** Conducted for 33k iterations with a learning rate of 1×10^{-5} .
- **Stage 2:** Conducted for 66k iterations with a learning rate of 2×10^{-6} .

Both stages utilize a mini-batch size of 16 (2 per GPU) and are performed on 8 NVIDIA A100 GPUs. Furthermore, in our confidence-aware adaptive noise injection strategy, we define the noise intensity parameters σ_{low} and σ_{high} to correspond to the confidence levels of the robotic arm’s state. Specifically, σ_{low} is set to 1.0 for regions with the lowest confidence, while σ_{high} is set to 0.85 for regions with the highest confidence.

C Supplementary Results

As shown in Fig. 5, Fig. 6, Fig. 7, and Fig. 8, we provide more experimental results and additional visual comparisons with the baseline method. These results highlight that our method achieves state-of-the-art quality, characterized by fine-grained interaction details, high visual fidelity, and robust spatiotemporal consistency.

D Limitations and Future Works

While our approach demonstrates significant potential, several challenges remain. First, the inherent constraints of the generative model lead to degraded performance in scenes involving substantial viewpoint variations. Second, the current 4D generative paradigm relies heavily on temporal continuity; when simulating a fixed-view camera for downstream tasks, the model must synthesize the target viewpoint directly rather than transitioning smoothly from the initial frame. This spatiotemporal discontinuity often compromises generation quality. Finally, inference efficiency remains a bottleneck, as generating a sequence of 49 frames currently requires approximately two minutes, failing to meet the requirements for real-time deployment.

Future research should prioritize optimizing performance under extreme viewpoint shifts and fixed viewpoints while improving computational efficiency. Regarding model training, beyond the multi-view fine-tuning employed here, future work could incorporate large-scale multi-view data during pre-training. This would endow the Vision-Language-Action (VLA) model with superior cross-viewpoint generalization, enabling advanced spatial reasoning and manipulation planning capabilities for Embodied AI.

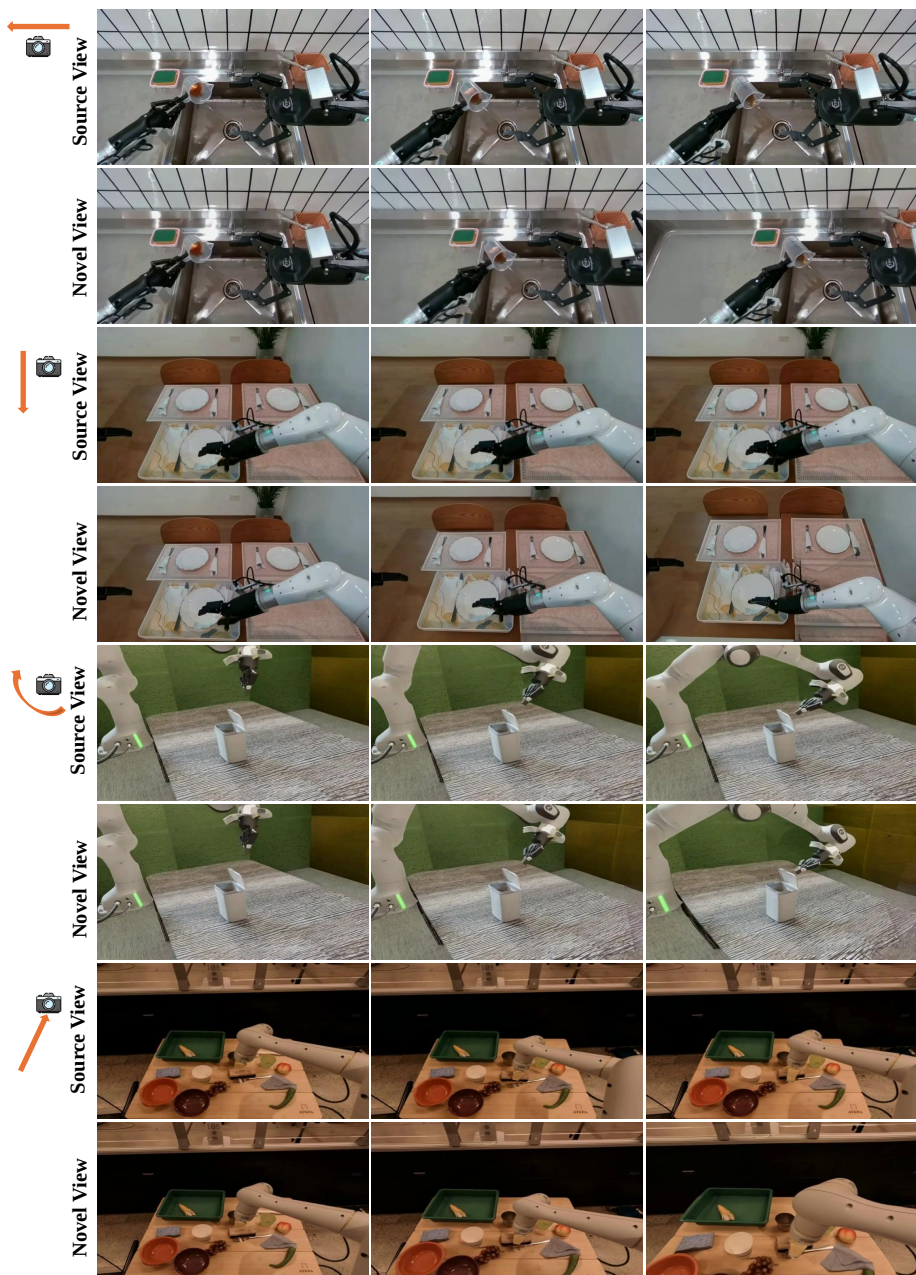


Fig. 5: Visualization results of Embodiment4D

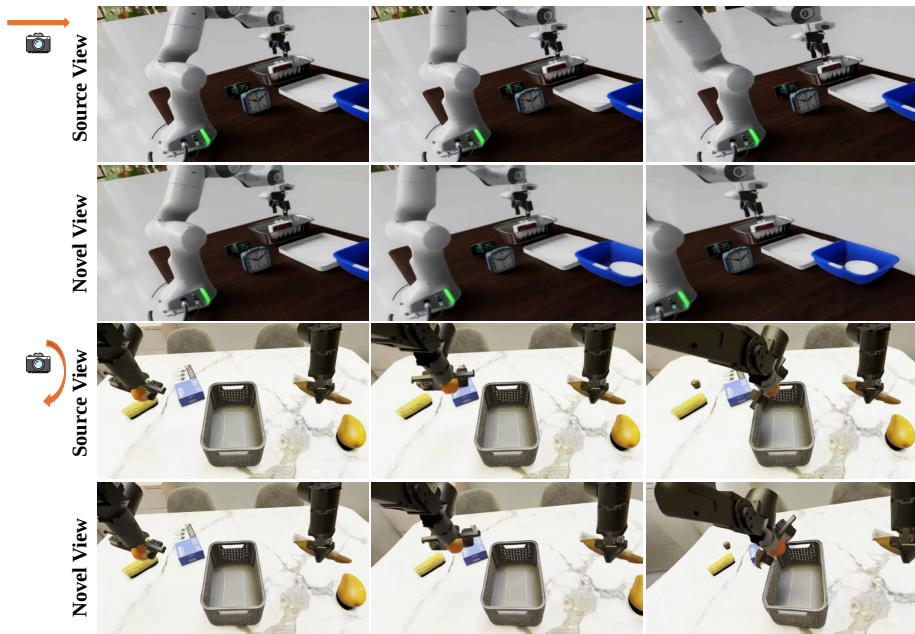


Fig. 6: Visualization results of Embody4D

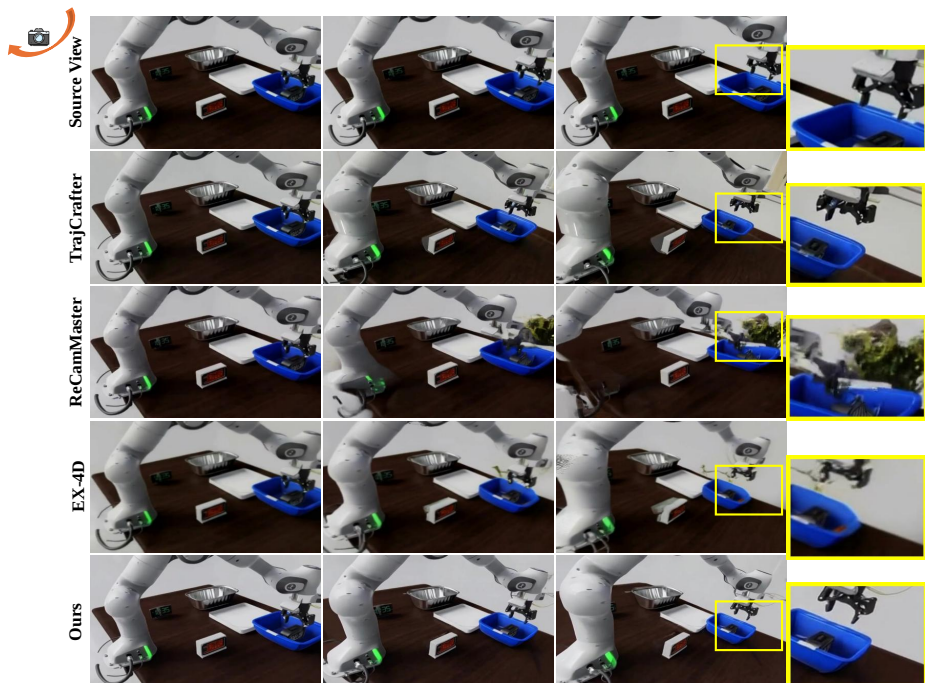


Fig. 7: Qualitative visualization results of our method and the baselines.

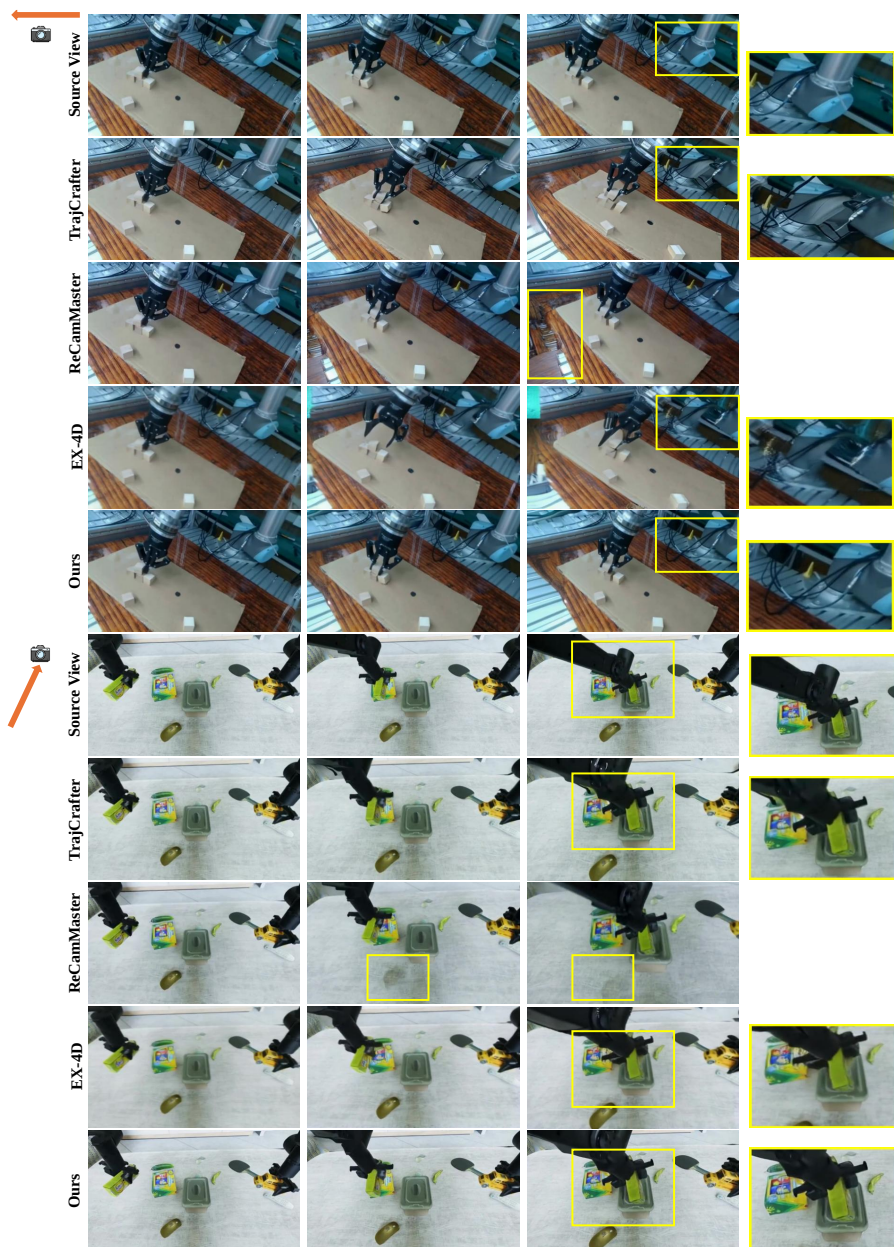


Fig. 8: Qualitative visualization results of our method and the baselines.

References

1. Achiam, J., Adler, S., Agarwal, S., Ahmad, L., Akkaya, I., Aleman, F.L., Almeida, D., Altenschmidt, J., Altman, S., Anadkat, S., et al.: Gpt-4 technical report. arXiv preprint arXiv:2303.08774 (2023)
2. Bai, J., Xia, M., Fu, X., Wang, X., Mu, L., Cao, J., Liu, Z., Hu, H., Bai, X., Wan, P., et al.: Recammaster: Camera-controlled generative rendering from a single video. arXiv preprint arXiv:2503.11647 (2025)
3. Carion, N., Gustafson, L., Hu, Y.T., Debnath, S., Hu, R., Suris, D., Ryali, C., Alwala, K.V., Khedr, H., Huang, A., et al.: Sam 3: Segment anything with concepts. arXiv preprint arXiv:2511.16719 (2025)
4. Dong, Q., Fu, Y.: Memflow: Optical flow estimation and prediction with memory. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 19068–19078 (2024)
5. Ling, L., Sheng, Y., Tu, Z., Zhao, W., Xin, C., Wan, K., Yu, L., Guo, Q., Yu, Z., Lu, Y., et al.: D13dv-10k: A large-scale scene dataset for deep learning-based 3d vision. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 22160–22169 (2024)
6. Todorov, E., Erez, T., Tassa, Y.: Mujoco: A physics engine for model-based control. In: 2012 IEEE/RSJ international conference on intelligent robots and systems. pp. 5026–5033. IEEE (2012)
7. Wang, J., Chen, M., Karaev, N., Vedaldi, A., Rupprecht, C., Novotny, D.: Vggt: Visual geometry grounded transformer. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 5294–5306 (2025)
8. Yang, Z., Teng, J., Zheng, W., Ding, M., Huang, S., Xu, J., Yang, Y., Hong, W., Zhang, X., Feng, G., et al.: Cogvideox: Text-to-video diffusion models with an expert transformer. arXiv preprint arXiv:2408.06072 (2024)
9. YU, M., Hu, W., Xing, J., Shan, Y.: Trajectorycrafter: Redirecting camera trajectory for monocular videos via diffusion models. arXiv preprint arXiv:2503.05638 (2025)